Attentive-Adaptive Network for Hyperspectral Images Classification With Noisy Labels

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***Abstract*— With the development of deep neural networks, hyperspectral image (HSI) classification systems have achieved a significant improvement. These systems require numerous and accurately labeled hyperspectral data to be adequately trained. However, noisy labels are inherent in real-world hyperspectral systems, resulting in unreliable decisions. To handle noisy labels in hyperspectral classification, an end-to-end attentive-adaptive network (AAN) is proposed for robust HSI classification training. The goal is to build a classifier with strong generalization capabilities that can be applied to both clean and noisy training sets without explicit noise label pretreatment. Specifically, a spectral stem network with a nonadjacent shortcut is exploited initially to redistribute the sensitive layers for noisy labels to achieve robust spectral representation. Then, a group-shuffle attention module is proposed to capture the discriminative and robust spatial–spectral features in the presence of noisy labels. Finally, an adaptive noise-robust loss (ANRL) function is developed to fight against noisy labels by learning a parameter to balance the normalized cross entropy (NCE) and reverse cross entropy (RCE). Experimental results on three HSI benchmark datasets with simulated noisy labels demonstrate the effectiveness of AAN on HSI classification.**

***Index Terms*— Deep learning, hyperspectral images classification, noisy labels.**

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I. Introduction

HYPERSPECTRAL images contain rich spectral and spatial information, which can stimulate the understanding and application of complex remote-sensing scenes. Meanwhile, due to its high training efficiency and high accuracy classification results, hyperspectral image classification methods have been widely used in many fields, such as mineral detection [1], environmental monitoring [2], and agricultural analysis [3].

Machine-learning-based methods have achieved tremendous success in the hyperspectral image (HSI) classification. Early works, such as logistic regression (LR) [4], linear discriminant analysis (LDA) [5], multinomial logistic regression (MLR) [6], [7], and support vector machine-Markov random field (SVM-MRF) [8], [9], [10], concentrate on feature engineering and classifier training for HSI classification. Recently, deep-learning methods have achieved significant advancements in HSI classification due to the powerful representation of learning potentials. With the aid of the local feature extraction capability of convolutional neural networks (CNNs), 1-D [11], 2-D [12], and 3-D [13] CNNs are extensively exploited in spectral, spatial, and spatio-spectral domains for HSI classification. From the perspective of sequential data, long short-term memory (LSTM) [14] and gate recurrent unit (GRU) [15] also demonstrate that recurrent neural network (RNN) is an effective solution for HSI classification. By resolving the long-distance feature dependencies, Transformer has recently been a hot topic in HSI classification, showcasing its superiority in feature extraction [16], [17]. However, most of the existing deep methods rely on the availability of large amounts of data that have been accurately annotated. When labels in training data are corrupted, the performance of the classifier degrades dramatically as the number of noisy labels increases.

Noisy labels are inevitable in real-world remote-sensing scenarios, that is, imprecise geolocalization or unenforceable ground investigations. Besides, large-scale and accurate labeling is a labor-intensive and cost-consuming task. In particular, the low interclass and high intraclass variabilities of hyperspectral images make labeling a particularly challenging assignment, even for seasoned experts. The issue of the noisy label is particularly acute in deep models due to the requirement of massive parameters to learn any complicated function. The overparameterization makes deep models vulnerable to memorizing any set of random noise [18]. The brilliant fitting ability may readily fit an entire training set

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regardless of the proportion of corrupted labels [19], however, leading to poor generalizability on a test set. As a result, robust deep models with high generalizability are required for hyperspectral classification in the presence of noisy labels.

To eliminate the negative influence of mislabeling, a straightforward way is to purify the training data by excluding the samples that are likely to be corrupted. Kang et al. [20] exploit a constrained energy minimization strategy on the extracted high separability spatial–spectral features to detect and correct the noisy labels. Tu et al. [21], [22] propose to identify the mislabeled training samples based on a density peak clustering algorithm, where false-labeled training samples are excluded by a local density-based decision strategy. However, the data cleaning approaches suffer from the over-cleaning issue, which is susceptible to resulting in the removal of precise true-labeled samples. Therefore, it is desirable to develop noise-tolerant deep network architectures to overcome the noisy labels for HSI classification.

Deep models are more prone to overfitting and memorizing corrupted labels [23] than traditional classifiers. Deep models are trained in a serial modular manner with two phases, where different layers are dedicated to fitting distinct kinds of information in the function space [24]. Generally, the first phase learns to fully represent the input data with low-level features. Then, the following phase is to view learning as data compression by forgetting the irrelevant details [25]. The interlocking of different layers increases the risk of overfitting the corrupted label. Given these considerations, Hamida et al. [26] utilize depthwise 3-D convolution to extract joint spectral and spatial information, which has been shown to be effective in dealing with noisy labels to some extent. Roy et al. [27] develop a lightweight heterogeneous kernel convolution to enhance the robustness of extracted spatial–spectral features under noisy training samples. However, robustness gains come at the expense of decreasing the representation ability of high-level semantic features. In the presence of clean training data, the performance of the model is compromised. It is difficult to forecast the presence of noisy labels in most practical HSI applications. How to give attention to the clean and noisy training set at the same time remains a difficult task that merits additional exploration.

The attention mechanism is one of the most prominent ideas in the deep-learning community, which is exploited to concentrate on the most relevant stimuli while fading out the irrelevant ones. Various attention mechanisms are employed in classification tasks due to the exceptional performances in HSI classification. Sun et al. [28] introduce the nonlocal attention in the spatial domain to suppress the effects of interfering pixels for discriminative spectral–spatial features extraction. Zhu et al. [29] employ a convolutional block attention module (CBAM) to adaptively select informative spectral and spatial information sequentially for HSI classification. Li et al. [30] conduct global context (GC) attention on the channel and position branches, respectively, to capture adaptive nonlinear feature relationships in the spectral and spatial dimensions. The attention-based methods achieve impressive results in classification accuracy when the training set is totally clean. Nevertheless, the attention mechanism will accelerate the performance degradation when the noisy labels are mixed in the training set due to the intensified data compression compared to the vanilla deep model. It is still a tough area to adapt the attention mechanism to a noise-robust deep architecture when noisy labels are present.

Categorical cross-entropy (CCE) is a standard configuration in deep classification models that exhibits great performance on correctly labeled data. However, CCE is susceptible to memorization effects [23], with which deep models lead to serious overfitting over noisy labels [31]. To mitigate this issue, there has been a surge of interest in attempting to exploit robust loss functions to train deep models against noisy labels. With the help of mutually boosted reverse CE (RCE) loss and normalized CE (NCE), a dual-channel network that exploits intermediate spatial and spectral information is proposed to reduce the influence of noisy labels for HSI classification [32]. Likewise, RCE and NCE are also adopted by Roy as the noise-tolerance objective function to overcome the noisy labels for HSI classification [27]. However, RCE and NCE losses are merged equally to accommodate a wide variety of data. A flexible and adaptive manner to balance the relationship between NCE and RCE may be beneficial in boosting classification performance.

In this article, an end-to-end attentive-adaptive network (AAN) is proposed for HSI classification without explicit preliminary treatment of the noise labels. We aim to enhance the effectiveness and robustness of CNNs for both clean and noisy training sets, which generate discriminative spatial–spectral features by alleviating the concentration of feature representation under the attention mechanism. A spectral stem network with a nonadjacent layer shortcut is exploited first to handle the highly correlated spectral signatures, where the shortcut is used to redistribute the sensitive layers to noisy labels for robust spectral representation. Then, a group-shuffle attention module is proposed to capture the discriminative spatial–spectral features in the presence of noisy labels. In the group-shuffle attention module, spectral and spatial attention are used to obtain sophisticated spectral–spatial features, while group and shuffle operations are used to alleviate the adverse effect of noisy labels due to the overrefined features brought about by the attention mechanism. Finally, a noise tolerance loss function is designed to fight against label noise by learning a parameter to balance the NCE and RCE, aiming to accommodate a variety of training sets adaptively.

The main contributions of this work are summarized as follows.

1) An end-to-end AAN is proposed for HSI classification in the presence of noisy labels. To the best of our knowledge, the proposed AAN is the first attention-aware noise-robust architecture that incorporates discriminative spatial–spectral information on the entire HSI training set containing clean and noisy labels simultaneously.
2) Group-shuffle operation is exploited to accommodate the attention mechanism to a noise-robust deep architecture when noisy labels are present. The spatial and spectral attention help to improve the spatial–spectral

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representation of interests, while the group-shuffle operation is carried out to alleviate the interlocking of deep models to combat the noisy labels.

3) An adaptive noise-robust loss (ANRL) function is designed to boost the performance on HSI classification when noisy land-cover is likely to occur in the training data. With a learned balance parameter, the loss function allows an adjustment in the line from RCE to NCE, showing broad adaptability for varied ratios of noisy labels.
4) Extensive experiments are conducted on the three available HSI datasets to evaluate AAN's performance. Experimental results demonstrate that the proposed method offers the state of the art accuracy when fighting against noisy labels.

The rest of the article is structured as follows. Section II describes some related work. Section III describes the proposed method. Section IV presents the relevant experimental results and highlights the comparison of our results with other results. Section VI concludes this work.

II. Related Work

A. Hyperspectral Image Classification

With the advancement of deep learning, many effective deep-learning models have been developed for HSI classification, which can be divided into spectral, spatial, and spatial–spectral approaches. For spectral-based methods, Hu et al. [11] proposed a 1-D CNN for HSI classification, which handled the spectral signature as a 1-D signal. Hong et al. [16] proposed the SpectralFormer, a Transformer-based backbone for spectral information extraction. However, the spectral-based methods use spectral information only, which ignores the rich spatial correlations. By exploiting the spatial information, a pixel-centric 3-D patch is used as the input for HSI classification. For example, Zhang et al. [33] proposed a local spatial sequential method, which preserves the local spatial information and integrates the features of nonlocal similar samples. Duan et al. [34] fused the dual spatial information by exploiting preprocessing feature extraction and postprocessing spatial optimization simultaneously. Recently, spatial–spectral methods have shown their effectiveness by exploiting spatial and spectral information synchronously. Li et al. [35] proposed a 3-D CNN to exploit spatial–spectral features directly from the HSI cubes. Sun et al. [17] proposed a spectral–spatial feature tokenization transformer to capture spectral–spatial features for HSI classification. However, these deep methods rely on large amounts of data that have been accurately annotated. They had limited capability to deal with the noisy labels.

B. Hyperspectral Images Classification With Noisy Labels

In recent years, in the field of remote-sensing image classification, many methods have been proposed to solve the problem of hyperspectral image classification with noisy labels. Tu et al. [36] proposed a method based on density peak noise label detection (DPNLD) to correctly classify mislabeled samples with higher accuracy, which can effectively discover and correct noisy labels to a certain extent. Jiang et al. [37] developed a random label propagation algorithm (RLPA) with superpixel constraints, which cleans up samples with noisy labels, thereby increasing the proportion of correct samples during training. However, RLPA is susceptible to the improper number of superpixels. Jiang et al. [38] further proposed multiscale segmentation-based multilayer spectral–spatial graphs (MSSGs) to regularize the cleansing of noisy labels. Tu et al. [21], [22], [39] proposed several different methods, such as spatial density clustering (SDP) and its extension (K-SDP), to detect noisy labels by spectral–spatial metric and obtain better results. All of the methods mentioned above are noise-cleaning methods that have achieved impressive results in terms of classification accuracy, but in the process of cleaning noisy labels, the useful spatial and spectral information in the noisy samples is ignored, which is also helpful for HSI classification. From the above consideration, a dual-channel residual network (DCRN) for hyperspectral image classification with noisy labels is proposed by Xu et al. [32], which designs robust spatial and spectral branches to fight against noisy labels. In addition, a robust loss function that combines RCE and NCE equally is also helpful to improve the accuracy in the presence of noisy labels. Nevertheless, DCRN treats spatial and spectral branches separately, where the spatial and spectral information are not utilized to the best advantage. A spatial–spectral method should be tried to fight against noisy labels. The relationship between RCE and NCE should also be investigated to adapt to different proportions of noisy labels. Our work originated from two aspects.

III. Proposed AAN Framework

A. Problem Formulation

Given the HSI, it is represented as $\mathbb{H} \in \mathbb{R}^{H\times W\times B}$, where H and W are the height and width of the spatial dimensions, respectively, and B is the channel of the spectral dimension. To fully exploit the spectral and spatial information, each pixel in $\mathbb{H}$ can be denoted as a spatial–spectral cube $x_i \in \mathbb{R}^{S\times S\times B}$, where S is the patch size of the cubes. Suppose that there are N labeled pixels $X = \{x_1, x_2, \ldots, x_N\}$, $Y = \{y_1, y_2, \ldots, y_N\} \in \mathbb{R}^{1\times 1\times C}$ is the corresponding set of one-hot label vectors, with C being the number of land-covers. The classical supervised HSI classification task assumes all the labels y_i are properly annotated. Unfortunately, the noisy label is inevitable in practical HSI classification. Consequently, the training dataset can be split into two parts $D = \{D_{\text{clean}}, D_{\text{noisy}}\}$, where $D_{\text{clean}} = \{(x_1, y_1), (x_2, y_2), \ldots, (x_M, y_M)\}$ is the noise-free dataset and $D_{\text{noisy}} = \{(x_{M+1}, \hat{y}_{M+1}), (x_{M+2}, \hat{y}_{M+2}), \ldots, (x_N, \hat{y}_N)\}$ is the noisy counterpart with $\hat{y}$ the noisy labels, and M is the number of believable labels. The task is to construct a noise-tolerant model on D to find out the function that can predict the true land covers of the rest of $HW - N$ samples even without observing which label is noisy in advance.

B. Overview

The motivation of this article is to establish an end-to-end deep model for HSI classification with noisy labels. Unlike DCRN [32], which combines separate spectral and

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spatial branches, our strategy is to extract discriminative and robust spectral–spatial features against noisy labels. To clearly demonstrate the proposed AAN, the overall framework is shown in Fig. 1, which is organized into three essential modules. **(a) Spectral feature refinement.** The correlated spectral signatures are refined initially, with residual connections introduced to improve robustness. **(b) Spectral–spatial aggregation with group-shuffle attention.** Sophisticated spectral–spatial features of interests are acquired, with the group and shuffle attentions to mitigate the negative effect of noisy labels. **(c) Adaptive noise-robust loss.** An ANRL function is developed to adjust NCE and RCE adaptively, with a learned balancing parameter to improve noise tolerance for different ratios of noisy labels.

C. Spectral Feature Refinement

A set of 3-D cubes $X = \{x_1, x_2, \ldots, x_N\} \in \mathbb{R}^{7\times7\times176}$ is used as the HSI data, while $Y = \{y_1, y_2, \ldots, y_N\} \in \mathbb{R}^{1\times1\times13}$ is the corresponding set of label vectors. To handle the high-dimensional spectral signatures, the spectral feature refinement module act as dimensionality reduction, which provides similar functionality as principal component analysis (PCA). In contrast to PCA, this module benefits the proposed AAN benefits from this module's end-to-end training with other modules. Specifically, three consecutive convolutional layers are used as the stem network to refine the spectral features. $1 \times 1 \times 7$ kernels are used throughout the three convolutional layers to prevent convolution operations from transforming the original spatial relationship. The strides are (1, 1, 2), (1, 1, 1), and (1, 1, 1), respectively. Twenty-four kernels are used in all three convolutional layers. Following each convolution operation, a batch normalization (BN) layer and a Mish activation function are performed. To redistribute the sensitive layers for robust spectral feature refinement, a nonadjacent layer residual block is carried out to connect the first and third feature maps. The residual connection offers an identity path that preserves spectral information that would have been abstracted through convolution layers. The slight improvement in spectral feature representation alleviates the accumulation of incorrect signals when noisy labels are encountered. Finally, a feature map of 24 $7 \times 7 \times 85$ is generated as the refined spectral features to overcome the noisy labels.

D. Spectral–Spatial Aggregation With Group-Shuffle Attention

The attention mechanism has been proven to be effective in HSI classification by selectively emphasizing informative channels or suppressing the influence of interfering pixels. The rules really work fine when the training set is properly labeled. The accuracy drops whenever noisy labels appear in the training set due to the overcentralized features brought by the attention mechanism. In this part, we introduce group-shuffle operation to accommodate the attention mechanism for robust spatial–spectral representation.

1) Spectral Feature Grouping: With the newly generated spectral feature map, 128 $1 \times 1 \times 85$ kernels are performed to obtain 3-D patches $f_i \in \mathbb{R}^{7\times7\times128}, i = 1, 2, \ldots, N$. Then, f_i is divided into G groups along the channel dimension, that is, $f_i = [f_{i1}, f_{i2}, \ldots, f_{iG}]$, $f_{ig} \in \mathbb{R}^{7\times7\times128/G}, g = 1, \ldots, G$. Each f_{ig} can be regarded as a subfeature. Double-branch attention is put forward to each subfeature to extract spectral and spatial features separately. The close-coupled modules would render the learned features too abstract to overfit in the presence of noisy labels. Different from the traditional double-branch strategy that shares the same input, we split the input features f_{ig} along the channel dimension into two branches

$$f_{ig} = \left[f_{ig}^t, f_{ig}^b\right] \tag{1}$$

where $f_{ig}^t \in \mathbb{R}^{7\times7\times64/G}$ and $f_{ig}^b \in \mathbb{R}^{7\times7\times64/G}$ are the nonoverlapping top and bottom branches.

The nonoverlapping f_{ig}^t and f_{ig}^b are prepared for spectral and spatial features, respectively, to reduce the interlocking of enhanced features. The incompact feature representation compensates for the deficiency of deep models in the presence of noisy labels. The top branch features are enhanced by channel attention to strengthening the spectral correlation between channels. In contrast, the bottom branch features are boosted by spatial attention to selectively highlight the informative pixels among the neighborhoods. The channel partition manner reduces the concentrated feature representation ability to enhance the generalization over noisy labels.

2) Channel Attention: As shown in Fig. 2, channel attention works on f_{ig}^t, which is exploited to model the complex correlations among channels. Global average pooling is first performed on f_{ig}^t to get a $64/G$ vector for channel representation. Then, a two-layer multilayer perceptron (MLP) is established to enhance the interrelationships among channels. Through the MLP, the information is exchanged among the channels with a channel reduction ratio r (e.g., 4) and then recover to the original number of channels. The channel attention coefficient A_{cha} can be obtained as

$$A_{\text{cha}} = \sigma\left(\left(\mathbf{W_1} \cdot \left(\mathbf{W_0} \cdot \text{GAP}\left(f_{ig}^t\right) + \mathbf{b_0}\right) + \mathbf{b_1}\right)\right) \tag{2}$$

where $\text{GAP}(\dot{})$ is the global average pooling, W_0 and b_0 are the learned parameter of the first layer, W_1 and b_1 are the learned parameters of the second layer, and $\sigma(\cdot)$ is the sigmoid activation function. Therefore, the enhanced channel feature for f_{ig} is written as

$$f_{ig}^{t'} = A_{\text{cha}} \cdot f_{ig}^t. \tag{3}$$

3) Spatial Attention: As shown in Fig. 2, spatial attention works on f_{ig}^B, which plays a complementary role to channel attention. Group normalization [40], [41] is conducted on f_{ig}^B for spatial-wise statistics. Then, the spatial attention coefficient A_{spa} is obtained by

$$A_{\text{spa}} = \sigma\left(W_2 \cdot \text{GN}\left(f_{ig}^b\right) + b_2\right) \tag{4}$$

where $\text{GN}(\dot{})$ is the group normalization, W_2 and b_2 are learned parameters of shape $\mathbb{R}^{64/G}$ for fully connection layer, and $\sigma(\cdot)$ is the sigmoid activation function. Consequently, the enhanced spatial features for f_{ig} is written as

$$f_{ig}^{b'} = A_{\text{spa}} \cdot f_{ig}^b. \tag{5}$$

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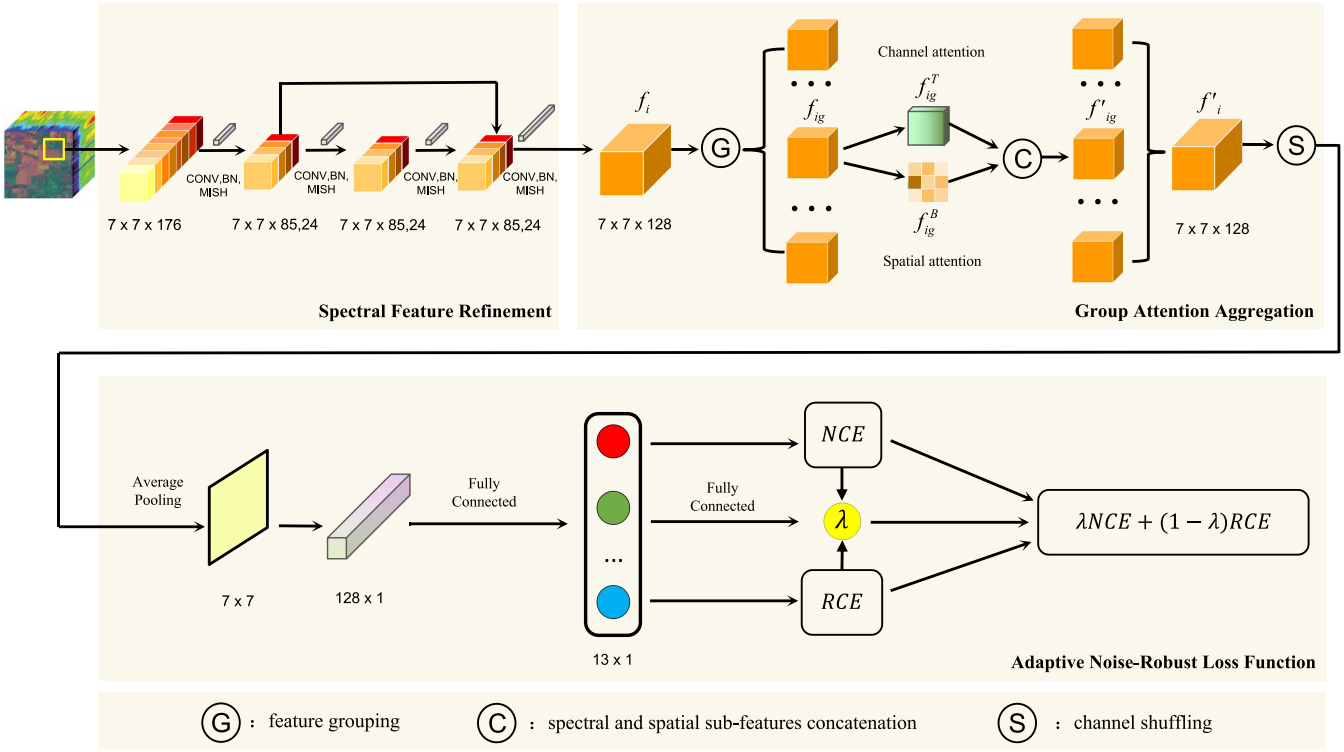

Fig. 1. Flowchart of the proposed AAN. A spectral stem network with a nonadjacent layer residual connection is first performed to redistribute the sensitive layers for robust spectral feature refinement. Then, a group-shuffle attention module is proposed to capture the spatial–spectral features of interest in scattered manners, aiming to alleviate the overrefined features when they encounter noisy labels. Finally, an ANRL function is developed to balance the NCE and RCE with the help of a fully connected layer to learn the balance parameter.

The two branches are then concatenated to generate the boosted spectral–spatial subfeatures $f'_{ig} = [f^{t'}_{ig}, f^{b'}_{ig}] \in \mathbb{R}^{7\times7\times128/G}$, which has the same number of channels as the input f_{ig}. It is worth noting that the proposed attention is a quite lightweight module compared with the traditional attention due to the multilevel channel division.

4) Subfeatures Aggregation: The G subfeatures are aggregated to obtain a feature of $\mathbb{R}^{7\times7\times128}$ by channel concatenation. Subsequently, a channel shuffle operation is performed to establish the cross-subfeatures correlation that flows along the channel dimension. The channel shuffle operation first reshapes the 128-dimensional features into $(7, 7, G, 128/G)$. The reshaped feature is then transposed to $(7, 7, 128/G, G)$ and finally flatten back to $(7 \times 7 \times 128)$. Through the channel shuffle operation, the final spectral–spatial features $f'_i \in \mathbb{R}^{7\times7\times128}$ are obtained. Then, the global average pooling layer with 7×7 kernel is employed to get a 128-D feature vector. The global average pooling layer aggregates the spectral–spatial information and reduces the number of parameters. The feature vector is fed to the softmax classifier and generates a classification prediction $\hat{y}_i \in \mathbb{R}^C$.

E. ANRL Function

Robust loss functions have been shown to be useful in achieving a low risk for unseen clean data even when noisy labels are present in the training data. Combining two mutually boosted robust loss functions has become mainstream in robust training [42]. However, it may not be easy to determine which loss function is important due to the unpredictable situation of noisy labels in the training set. In this article, this is achieved by devising an adaptive noise-robust function based on the well-known NCE and RCE loss functions.

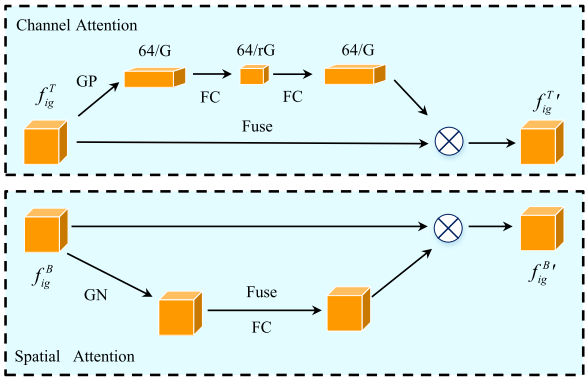

Fig. 2. Channel and spatial attentions.

For a sample x, the probability output of a classifier is denoted as $p(c \mid x)$, where $\sum_{c=1}^{C} p(c \mid x) = 1$. Correspondingly, the real distribution over different labels for sample x is $q(c \mid x)$ and $\sum_{c=1}^{C} q(c \mid x) = 1$. RCE is the reverse version of CE, which switches the order of real distribution q and predicted probability p. It lowers the likelihood of at least one incorrect label

$$\mathrm{RCE}(p, q) = -\sum_{c=1}^{C} p(c \mid x) \log q(c \mid x). \tag{6}$$

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NCE is the normalized CE that maximizes the probability of the correct labels

$$\mathrm{NCE}(p,q)=\frac{-\sum_{c=1}^{C} q(c\mid x)\log p(c\mid x)}{-\sum_{j=1}^{C}\sum_{c=1}^{C} q(y=j\mid x)\log p(y=j\mid x)}. \tag{7}$$

NCE and RCE are robust loss functions designed from different perspectives. To balance the relationship of NCE and RCE, a fully connected layer is exploited on the classifier's output to learn a balance parameter λ. Therefore, the ANRL function is written as

$$\mathrm{ANRL}(p,q)=\lambda\,\mathrm{NCE}(p,q)+(1-\lambda)\,\mathrm{RCE}(p,q) \tag{8}$$

where $\lambda=\mathrm{FC}(p)$, and FC() is an additional fully-connected layer. The advantage of the adaptive λ is that we do not need to reset the parameters when the noise ratios are unpredictable.

IV. Experimental Results and Analysis

In this section, detailed experiments will be conducted to demonstrate the effectiveness of attention-adaptive networks on the HSI classification task. Specifically, the proposed AAN has experimented on the Kennedy Space Center (KSC), University of Pavia (UP), and Salinas datasets. To better evaluate the performance of the proposed method, three commonly used metrics are evaluated, namely overall classification accuracy (OA), average classification accuracy (AA), and Kappa coefficient. All experiments were conducted with ten runs by using randomly selected training data.

A. Experimental Dataset

Three publicly available hyperspectral image databases, that is, KSC, UP, and Salinas, are used in the experiments to evaluate the proposed AAN method in the presence of noisy labels.

1) KSC: The KSC dataset was acquired by NASA AVIRIS in 1996 over KSC, FL, USA. The image of the KSC scene was gathered with wavelengths of $400-2500$ nm. The images have a spatial dimension of 512×614 pixels, and a spectral depth of 176 bands after a few low signal-to-noise ratios (SNRs) bands are discarded. For classification purposes, the KSC dataset consists of 5202 samples from 13 different bands representing the various land cover types present in the environment defined for the site categories.

2) UP: The UP dataset was gathered by the ROSIS sensor during a flight campaign over Pavia, northern Italy. After removing noisy bands, the image in the UP scene contains 103 spectral bands in the wavelength range of $430-860$ nm with the spatial dimension of 610×340 pixels. The samples contained nine classes of interest, most of which are crops in different growth phases.

3) SV: The SV dataset was acquired over Salinas Valley, California, by the Airborne Visible/Infrared Imaging Spectrometer sensor and is characterized by high spatial resolution (3.7 m per pixel). The image consists of 512 lines with 217 samples. There were 224 spectral reflectance bands in Salinas Valley ranging from 400 to 2500 nm. After removing 20 water absorption bands, the final 204 bands were retained in the experiments. The available category covered 16 classes of interest.

B. Experimental Settings

The proposed AAN is evaluated on the public datasets, that is, KSC, UP, and Salinas Valley (SV) datasets with simulated noisy labels. Noisy labels are generated by randomly replacing the real labels with other corrupted labels. The AAN is implemented in PyTorch 1.8.0, GPU with CUDA 11.1.1 for experimental training. Following DCRN [32], the batch size is set to 16 due to the small training set. The ADAM optimizer is used to control the training process with a learning rate of 0.001.

C. Influence of ANRL Function

The noise-robust loss function is crucial to prevent overfitting for the classifier learned with noisy and noise-free data. In this part, we evaluate the proposed ANRL function on the KSC dataset with 20 clean samples and varied numbers of mislabeled samples (ranging from 10 to 40). As shown in Table I, we can observe that ANRL improves the quality of a reliable model in the presence of noisy labels. Although CE is the most commonly used loss function in deep models, it has poor robustness to noisy labels. Therefore, CE achieves the worst performance in all experiments. NCE is the normalized cross-entropy loss function, and RCE is the reverse cross-entropy loss function. Compared with the CE loss function, the two loss functions improve the robustness of the model to some extent. However, they are only partially robust in the presence of noisy labels. NRL combines the advantages of NCE and RCE, which achieves better performances. However, the simple average strategy ignores the asymmetric relationship between NCE and RCE. Benefiting from adaptive learning, ANRL learns the parameter λ to balance the proportion of NCE and RCE adaptively. The learned λ improves classification accuracy when compared to NRL, demonstrating broad adaptability on all different proportions of noisy labels. The learned values of λ are 0.54, 0.53, 0.51, and 0.48 when the number of noisy labels is 10, 20, 30, and 40, respectively. Overall, the importance of the RCE increases as the number of noisy labels grows.

D. Noise Robustness Analysis of AAN

The well-designed network structure is also a key factor to gain strong robustness to noisy labels. In this part, we compare the proposed AAN with several networks, including the 3-D CNN [35], spectral–spatial residual network (SSRN) [13], and DCRN [32]. For a fair comparison, CE is used as the loss function in all networks. Experiments are conducted on KSC with 20 clean samples and varied numbers (10–40) of noisy samples to evaluate the robustness of compared networks. The experimental results are reported in Table II. From Table II, we can observe that AAN achieves the highest experimental results in all experimental settings. It demonstrates that the proposed

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TABLE I

CLASSIFICATION ACCURACY FOR THE KSC DATASET. CLASSIFICATION ACCURACY OBTAINED BY AAN WITH CE LOSS, AAN WITH NCE LOSS, AAN WITH RCE LOSS, AAN WITH NRL FUNCTION, AND AAN WITH ANRL FUNCTION WITH 20 TRUE SAMPLES AND DIFFERENT NUMBER OF MISLABELED SAMPLES

class	The number of true and noisy training samples(KSC)																			
	20(clean)+10(noisy)					20(clean)+20(noisy)					20(clean)+30(noisy)					20(clean)+40(noisy)				
	CE	NCE	RCE	NRL	ANRL	CE	NCE	RCE	NRL	ANRL	CE	NCE	RCE	NRL	ANRL	CE	NCE	RCE	NRL	ANRL
1	88.83	96.05	**99.18**	97.35	97.80	83.76	92.58	**98.19**	97.47	97.64	86.74	92.71	96.80	**97.74**	97.66	84.65	87.86	90.20	**97.37**	96.55
2	91.21	93.50	93.30	93.09	**94.28**	83.08	84.60	88.58	89.87	**94.26**	67.35	85.76	89.07	86.65	**92.36**	61.14	76.68	91.16	**95.12**	83.83
3	96.74	92.71	89.24	96.86	**98.13**	90.38	70.72	90.26	**94.19**	91.09	89.24	70.87	95.57	81.66	**96.02**	88.23	63.07	92.15	88.73	**94.85**
4	80.24	61.91	**88.45**	87.96	87.20	71.71	59.70	67.41	80.00	**81.71**	64.03	40.89	68.38	**82.83**	74.93	56.65	42.57	56.53	73.56	**81.90**
5	87.82	77.66	**95.42**	93.07	91.28	80.28	65.47	**93.60**	88.30	91.85	83.62	60.00	85.42	79.40	**92.26**	81.92	76.75	**92.82**	85.57	87.40
6	78.42	93.37	97.88	**98.80**	98.68	89.28	95.28	95.24	95.65	**96.74**	77.66	**95.99**	92.15	94.52	90.64	68.59	82.96	**94.12**	90.53	90.93
7	97.79	82.40	**99.41**	98.90	93.91	91.21	87.33	**100.00**	99.16	98.49	95.84	81.59	91.92	**99.45**	92.14	84.42	79.82	**100.00**	98.54	95.82
8	88.88	93.43	97.17	96.38	**97.43**	86.72	88.14	**96.15**	90.65	95.73	75.86	85.76	96.18	89.90	**98.02**	72.20	77.39	89.25	**91.90**	86.28
9	90.35	91.37	93.59	**98.72**	98.31	87.94	86.88	94.10	**96.48**	93.65	78.23	81.81	85.63	**96.17**	91.72	71.97	75.20	**97.84**	90.70	87.46
10	83.58	96.22	96.74	97.97	**99.30**	79.73	92.71	96.12	**96.84**	96.82	62.68	92.65	89.08	**93.16**	90.14	62.18	84.76	87.38	87.33	**89.66**
11	91.36	98.36	**99.85**	98.52	99.34	88.21	94.68	96.73	**98.84**	98.72	84.68	94.09	95.97	94.76	**98.40**	80.91	96.57	98.07	96.63	**98.10**
12	84.13	89.10	95.03	94.61	**95.20**	74.13	90.69	87.75	91.61	**93.71**	75.95	86.50	91.31	90.11	**91.54**	71.19	83.77	85.90	86.21	**90.00**
13	98.26	99.99	100.00	100.00	**100.00**	97.51	99.95	100.00	100.00	**100.00**	90.77	98.21	100.00	100.00	**100.00**	87.91	97.70	99.93	97.71	**100.00**
OA	89.85	92.95	96.70	97.10	**97.39**	86.12	89.54	94.24	95.22	**95.83**	80.59	87.18	92.86	93.63	**94.61**	76.88	83.61	91.84	92.33	**92.72**
(%)	(3.59)	(1.29)	(0.95)	(1.30)	**(0.98)**	(3.93)	(2.35)	(3.01)	(1.39)	**(1.52)**	(3.57)	(3.24)	(3.69)	(2.31)	**(1.26)**	(4.28)	(3.43)	(1.40)	(1.93)	**(2.69)**
AA	89.05	89.70	95.79	96.33	**96.22**	84.92	85.29	92.63	93.77	**94.65**	79.43	82.06	90.58	91.26	**92.76**	74.77	78.86	90.41	90.76	**90.98**
(%)	(3.61)	(2.12)	(0.97)	(1.98)	**(1.73)**	(3.73)	(5.35)	(3.13)	(1.88)	**(1.99)**	(4.04)	(4.61)	(4.19)	(2.83)	**(1.30)**	(4.89)	(4.34)	(1.29)	(1.88)	**(3.08)**
KAPPA	0.89	0.92	0.96	0.97	**0.97**	0.85	0.88	0.94	0.95	**0.95**	0.78	0.86	0.92	0.93	**0.94**	0.74	0.82	0.91	0.91	**0.92**
	(0.04)	(0.04)	(0.01)	(0.01)	**(0.01)**	(0.04)	(0.02)	(0.03)	(0.01)	**(0.02)**	(0.04)	(0.04)	(0.04)	(0.03)	**(0.01)**	(0.05)	(0.04)	(0.02)	(0.02)	**(0.03)**

TABLE II

CLASSIFICATION ACCURACY FOR THE KSC DATASET. CLASSIFICATION ACCURACY OBTAINED BY THE 3-D CNN, SSRN, DCRN, AND AAN WITH COMMON CE LOSS, RESPECTIVELY, WITH 20 TRUE SAMPLES AND A DIFFERENT NUMBER OF MISLABELED SAMPLES

class	The number of true and noisy training samples(KSC)															
	20(clean)+10(noisy)				20(clean)+20(noisy)				20(clean)+30(noisy)				20(clean)+40(noisy)			
	CNN	SSRN	DCRN	ANN	CNN	SSRN	DCRN	ANN	CNN	SSRN	DCRN	ANN	CNN	SSRN	DCRN	ANN
1	59.34	73.42	84.28	**88.83**	48.03	65.92	77.84	**83.76**	34.37	50.65	78.33	**86.74**	30.46	44.03	74.60	**84.65**
2	52.23	75.36	79.95	**91.21**	38.51	55.75	76.12	**83.08**	28.24	41.73	65.56	**67.35**	21.06	38.97	59.87	**61.14**
3	49.40	77.96	88.95	**96.74**	37.13	55.33	78.81	**90.38**	30.99	42.06	74.84	**89.24**	25.99	40.83	81.65	**88.23**
4	25.32	61.99	70.16	**80.24**	19.33	46.06	66.70	**71.71**	16.98	35.54	63.76	**64.03**	17.01	22.15	50.60	**56.65**
5	41.77	69.62	**89.30**	87.82	28.49	60.34	**85.99**	80.28	22.80	47.97	75.97	**83.62**	20.72	43.23	66.41	**81.92**
6	42.12	73.07	**88.33**	78.42	30.40	52.90	66.52	**89.28**	22.50	39.79	**79.43**	77.66	19.34	38.11	**69.06**	68.59
7	67.58	75.89	96.85	**97.79**	48.41	64.92	89.62	**91.21**	47.43	45.72	87.81	**95.84**	32.31	36.72	83.69	**84.42**
8	44.83	79.09	87.24	**88.88**	32.10	60.44	79.86	**86.72**	27.46	39.84	66.35	**75.86**	22.53	36.12	65.87	**72.20**
9	62.14	77.94	87.73	**90.35**	43.38	55.80	77.63	**87.94**	37.14	49.23	**80.05**	78.23	34.89	37.77	**73.05**	71.97
10	51.22	67.80	74.97	**83.58**	35.32	62.49	66.45	**79.73**	26.13	45.45	**62.84**	62.68	20.58	35.62	**67.93**	62.18
11	66.74	79.55	**91.61**	91.36	51.98	66.74	84.25	**88.21**	44.15	49.81	74.17	**84.68**	32.17	47.89	**81.40**	80.91
12	44.25	79.21	78.93	**84.13**	30.05	61.95	70.60	**74.13**	25.57	41.75	58.20	**75.95**	22.55	38.67	64.86	**71.19**
13	81.27	94.57	95.52	**98.26**	60.82	87.03	96.10	**97.51**	58.06	64.70	**91.47**	90.77	44.66	59.26	84.17	**87.91**
OA	57.59	78.67	86.25	**89.85**	42.86	65.42	79.84	**86.12**	36.20	49.01	75.31	**80.59**	29.86	43.28	72.95	**76.88**
(%)	(2.98)	(5.19)	(4.94)	**(3.59)**	(4.03)	(9.11)	(7.23)	**(3.93)**	(2.99)	(4.30)	(7.59)	**(3.57)**	(2.63)	(9.86)	(10.53)	**(4.28)**
AA	52.94	75.81	85.68	**89.05**	38.76	61.21	78.19	**84.92**	32.45	45.71	73.75	**79.43**	26.48	39.95	71.01	**74.77**
(%)	(3.01)	(5.25)	(4.01)	**(3.61)**	(2.86)	(9.34)	(6.84)	**(3.73)**	(1.83)	(2.95)	(6.48)	**(4.04)**	(2.24)	(9.20)	(8.06)	**(4.89)**
KAPPA	0.53	0.76	0.85	**0.89**	0.37	0.62	0.78	**0.85**	0.30	0.44	0.72	**0.78**	0.23	0.37	0.70	**0.74**
	(0.03)	(0.06)	(0.05)	**(0.04)**	(0.04)	(0.10)	(0.08)	**(0.04)**	(0.03)	(0.04)	(0.08)	**(0.04)**	(0.03)	(0.11)	(0.12)	**(0.05)**

AAN is the most robust network to noisy labels among all compared methods. The 3-D CNN, SSRN, and ANN are all spectral–spatial-based HSI classification methods. The 3-D CNN performs naive 3D convolution to extract spectral–spatial features. However, it does nothing to address the negative influence of noisy labels in the training set. Therefore, the 3-D CNN performs worst in the presence of noisy labels. SSRN is an end-to-end SSRN, where residual blocks connect every other 3-D convolutional layer through identity mapping. Due to the residual connection, SSRN is more robust than the 3-D CNN, where residual blocks emphasize distinct features while suppressing ambiguous ones. Nevertheless, the residual connection is still not enough to manage the noisy labels, resulting in inferior precision. Compared to the 3-D CNN and SSRN, the proposed AAN alleviates the interlocking of networks by performing the group-shuffle strategy. Therefore, the dispersive network architecture helps the AAN achieve the best performance against noisy labels. DCRN [32] is also a noise-tolerant network, which proposes a dual way to resolve HSI classification with noisy labels by exploiting spatial and spectral separately. Different from DCRN, our AAN exploits spatial–spectral information with an attention mechanism, giving dual attention to the clean and noisy training set. Table III gives the results on the KSC dataset with 20 clean samples only. From Tables II and III, it can be seen that the proposed AAN achieves the best performance in all situations, demonstrating its robustness and adaptability when the labels in the training set are clean or noisy.

E. Influence of Residual Connection in Spectral Feature Refinement

The residual connection in spectral feature refinement is used to reduce overfitting and improve robustness. In this

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TABLE III

CLASSIFICATION ACCURACY FOR THE KSC DATASET WITH 20 TRUE SAMPLES WITHOUT NOISE SAMPLES. CLASSIFICATION ACCURACY OBTAINED BY THE 3-D CNN, SSRN, DCRN, AND AAN WITH CE LOSS

	CNN	SSRN	DCRN	AAN
OA	81.91	97.80	99.08	**99.22**
(%)	(4.71)	(0.58)	(0.48)	**(0.41)**
AA	77.73	97.28	98.33	**98.96**
(%)	(4.81)	(1.13)	(1.45)	**(0.34)**
KAPPA	0.80	0.98	0.99	**0.99**
	(0.05)	(0.01)	(0.01)	**(0.01)**

TABLE IV

CLASSIFICATION ACCURACY FOR THE KSC DATASET. CLASSIFICATION ACCURACY OBTAINED BY THE AAN WITHOUT RESIDUAL CONNECTION (W/O RESIDUAL) AND THE AAN WITH RESIDUAL CONNECTION (RESIDUAL) IN SPECTRAL FEATURE REFINEMENT WITH 20 CLEAN SAMPLES AND DIFFERENT NUMBER OF MISLABELED SAMPLES

Class	The number of true and noisy training samples(KSC)							
	20(clean)+10(noisy)		20(clean)+20(noisy)		20(clean)+30(noisy)		20(clean)+40(noisy)	
	w/o residual	residual	w/o residual	residual	w/o residual	residual	w/o residual	residual
OA	96.17	**97.39**	94.46	**95.83**	92.45	**94.61**	90.57	**92.72**
(%)	(1.22)	**(0.98)**	(1.85)	**(1.52)**	(2.85)	**(1.26)**	(3.64)	**(2.69)**
AA	95.14	**96.22**	93.07	**94.65**	90.70	**92.76**	89.06	**90.98**
(%)	(1.74)	**(1.73)**	(2.40)	**(1.99)**	(2.44)	**(1.30)**	(2.74)	**(2.08)**
KAPPA	0.96	**0.97**	0.94	**0.95**	0.92	**0.94**	0.89	**0.92**
	(0.01)	**(0.01)**	(0.02)	**(0.02)**	(0.03)	**(0.01)**	(0.04)	**(0.03)**

section, experiments are conducted to explore the effects of residual connection in the spectral feature refinement module on KSC with 20 clean samples and a varied number of noisy labels (10, 20, 30, and 40). The OAs, AAs, and KAPPAs of AAN are shown in Table IV. The residual connection in the spectral feature refinement module improves the performance of the proposed AAN by allowing information to flow more freely between layers. This pattern helps to alleviate the interlocking of sequential layers, avoiding the accumulation of incorrect signals from noisy labels.

F. Influence of Group-Shuffle Attention

The AAN incorporates the attention mechanism with group-shuffle operation into a noise-resistant deep architecture, with the goal of obtaining a model with strong generalization ability that is suited for practical HSI applications, regardless of whether noisy labels are present in the training set or not. To detail this experiment, we conduct some analysis about the attention mechanism on KSC with eight clean samples and different numbers of mislabeled samples (ranging from 4 to 12). To shed light on the contributions of the group-shuffle attention, Table V reports the classification results of AAN without attention or with self-attention on the KSC datasets. There are several observations provided in Table V. The experimental results show that it is effective to employ group-shuffle attention in the noise-robust model, where group-shuffle attention achieves the best performance in all noisy scenes. Contrary to expectations, the widely used self-attention performs the worst among the three methods in the presence of noisy labels, even lower than that without attention. This phenomenon highlights how an underutilized attention mechanism can have a negative impact on the model's robustness when dealing with noisy labels. The main reason is that traditional attention intensifies information compression, which increases the risk of overfitting corrupted labels. Instead, the group-shuffle operation alleviates the over-refined features caused by the attention mechanism to capture the discriminative spatial–spectral features in the presence of noisy labels.

Before the spatial and channel attention, each subfeature is decomposed into two branches along the channel dimension. Experiments have also been conducted to verify the advantage of the strategy that decomposes the input features into two branches. The aim is to alleviate the interlocking of deep models by reducing the number of channels for spatial and channel attention. The experiments are conducted on KSC with AAN ($G = 8$) for spatial and channel attention that performs on the partitioned channels and the shared all channels (AAN-a). The results in Table VI show that the partitioned channels are superior to the shared all channels for spatial and channel attention. The reason lies in that the robustness of the model is improved by relieving the feature compression ability of the attention mechanism in the presence of noisy labels. In addition, the computation cost is also reduced by performing on the half of channels for spatial and spectral attention, which obtains lightweight attention modules.

G. On the Number of Groups G

In the group-shuffle attention, the input features are first split into G subfeatures along the channel dimension. For the AAN, it was essential to set the number of groups G. In this part, we analyze the effect of the number of groups G by conducting experiments on the KSC dataset with eight clean samples and a different number of mislabeled samples (4, 8, and 12). Table VII shows the classification accuracy with different G. As can be seen in Table VII, the AAN obtains the highest classification accuracy when $G = 8$. The results are in line with the expectation that too large or too small number of groups leads to poor performance. A small number of groups G has a limited effect on alleviating the interlocking of spatial–spectral features in the presence of noisy labels. In contrast, a large number of groups G destroy the relationship among channels, which is not conducive to fitting on the clean samples.

H. On the Effect of Shuffle

Channel shuffle is performed to help information flow across the refined subfeature channels. Table VIII compares the performances of the AAN (group number is set to 8) with/without channel shuffle. It is clear that channel shuffle operation boosts classification accuracy by 1%–2% for different settings. This finding confirms that the shuffle unit improves model robustness, demonstrating the significance of cross-group information interchange against noisy labels.

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TABLE V

CLASSIFICATION ACCURACY FOR THE KSC DATASET WITH EIGHT CLEAN SAMPLES AND DIFFERENT NUMBER OF MISLABELED SAMPLES. CLASSIFICATION ACCURACY OBTAINED BY THE AAN WITHOUT ATTENTION, THE AAN WITH SELF-ATTENTION, AND THE AAN WITH GROUP-SHUFFLE ATTENTION, RESPECTIVELY

class	The number of true and noisy training samples(KSC)								
	8(clean)+4(noisy)			8(clean)+8(noisy)			8(clean)+12(noisy)		
	no-attention	self-attention	group-shuffle	no-attention	self-attention	group-shuffle	no-attention	self-attention	group-shuffle
1	**98.13**	93.97	97.36	91.89	69.00	**92.53**	95.58	70.84	**97.64**
2	**91.20**	71.95	89.15	81.37	68.09	**92.29**	68.39	54.04	**84.50**
3	**90.29**	68.89	90.11	**95.81**	47.14	83.89	86.08	51.85	**88.61**
4	62.72	23.47	**78.69**	48.99	31.06	**60.53**	52.92	24.83	**53.78**
5	**88.18**	49.97	85.27	81.44	46.52	**85.95**	**80.41**	32.95	76.85
6	**97.84**	56.57	89.05	85.28	37.01	**94.63**	81.92	18.75	**83.32**
7	94.58	94.73	**95.41**	**100.00**	56.81	97.04	**91.88**	35.27	88.21
8	82.50	40.47	**93.12**	**93.47**	41.79	77.85	79.79	35.67	**86.13**
9	**91.60**	76.46	91.53	85.25	66.80	**85.94**	74.80	59.78	**83.01**
10	88.04	68.02	**90.87**	89.16	55.77	**90.40**	80.10	67.64	**82.82**
11	**92.76**	82.56	92.68	94.21	90.43	**95.12**	**91.72**	83.98	82.61
12	85.60	62.97	**86.83**	81.91	59.97	**88.70**	75.41	34.40	**76.38**
13	100.00	77.21	**100.00**	89.81	85.60	**98.41**	94.10	74.93	**95.41**
OA	91.35	70.41	**92.65**	87.25	64.26	**89.45**	83.74	57.13	**85.95**
(%)	(1.53)	(8.04)	**(2.18)**	(4.40)	(9.58)	**(3.68)**	(3.86)	(11.16)	**(4.37)**
AA	89.50	66.71	**90.77**	86.04	58.15	**87.94**	81.01	49.61	**83.02**
(%)	(1.77)	(5.48)	**(2.97)**	(2.74)	(8.38)	**(3.38)**	(3.73)	(9.58)	**(4.04)**
KAPPA	0.90	0.67	**0.92**	0.86	0.60	**0.88**	0.82	0.52	**0.84**
	(0.02)	(0.09)	**(0.02)**	(0.05)	(0.10)	**(0.04)**	(0.04)	(0.12)	**(0.05)**

TABLE VI

CLASSIFICATION ACCURACY FOR THE KSC DATASET. CLASSIFICATION ACCURACY OBTAINED BY THE AAN WITH ALL (AAN-A) OR HALF CHANNELS (AAN) FOR SPATIAL AND CHANNEL ATTENTIONS WITH 20 CLEAN SAMPLES AND DIFFERENT NUMBER OF MISLABELED SAMPLES

Class	The number of true and noisy training samples(KSC)							
	20(clean)+10(noisy)		20(clean)+20(noisy)		20(clean)+30(noisy)		20(clean)+40(noisy)	
	AAN-a	AAN	AAN-a	AAN	AAN-a	AAN	AAN-a	AAN
1	97.22	**97.80**	97.40	**97.64**	95.37	**97.66**	**97.65**	96.55
2	**97.01**	94.28	90.35	**94.26**	89.37	**92.36**	**93.28**	83.83
3	90.33	**98.13**	**94.49**	91.09	89.75	**96.02**	86.18	**94.85**
4	81.76	**87.20**	71.27	**81.71**	58.43	**74.93**	66.94	**81.90**
5	90.32	**91.28**	83.16	**91.85**	90.43	**92.26**	84.96	**87.40**
6	98.58	**98.68**	**97.86**	96.74	**93.26**	90.64	90.61	**90.93**
7	**97.31**	93.91	**99.12**	98.49	**93.09**	92.14	**97.94**	95.82
8	96.31	**97.43**	89.63	**95.73**	90.82	**98.02**	**91.97**	86.28
9	**98.76**	98.31	**93.84**	93.65	90.56	**91.72**	**91.46**	87.46
10	98.08	**99.30**	96.11	**96.82**	**93.62**	90.14	88.35	**89.66**
11	**99.41**	99.34	98.05	**98.72**	**99.44**	98.40	96.12	**98.10**
12	94.48	**95.20**	93.27	**93.71**	89.20	**91.54**	83.63	**90.00**
13	100.00	**100.00**	100.00	**100.00**	100.00	**100.00**	95.64	**100.00**
OA	96.62	**97.39**	94.50	**95.83**	92.61	**94.61**	91.35	**92.72**
(%)	(1.45)	**(0.98)**	(2.70)	**(1.52)**	(1.89)	**(1.26)**	(2.75)	**(2.69)**
AA	95.35	**96.22**	92.66	**94.65**	90.26	**92.76**	89.60	**90.98**
(%)	(2.08)	**(1.73)**	(3.80)	**(1.99)**	(2.51)	**(1.30)**	(3.19)	**(3.08)**
KAPPA	0.96	**0.97**	0.94	**0.95**	0.92	**0.94**	0.90	**0.92**
	(0.02)	**(0.01)**	(0.03)	**(0.02)**	(0.02)	**(0.01)**	(0.03)	**(0.03)**

TABLE VII

CLASSIFICATION ACCURACY FOR THE KSC DATASET WITH EIGHT TRUE SAMPLES AND VARIED NUMBER OF MISLABELED SAMPLES. CLASSIFICATION ACCURACY OBTAINED BY DIVIDING THE FEATURES INTO 1, 2, 4, 8, AND 16 GROUPS, RESPECTIVELY

class	The number of true and noisy training samples(KSC)														
	8(clean)+4(noisy)					8(clean)+8(noisy)					8(clean)+12(noisy)				
	G=1	G=2	G=4	G=8	G=16	G=1	G=2	G=4	G=8	G=16	G=1	G=2	G=4	G=8	G=16
1	94.84	96.63	96.33	97.36	**97.91**	94.22	93.17	92.75	92.53	**94.82**	94.68	93.64	92.06	**97.64**	92.95
2	**91.99**	81.81	83.72	89.15	78.82	82.18	81.45	88.57	**92.29**	83.83	69.95	**84.72**	77.12	84.50	79.22
3	**97.14**	95.18	94.92	90.11	91.79	95.54	94.91	**96.63**	83.89	92.79	91.75	79.10	87.35	88.61	**94.13**
4	59.65	**81.57**	65.52	78.69	44.14	**72.26**	60.23	57.61	60.53	61.16	**65.26**	41.76	45.35	53.78	50.74
5	**90.05**	82.29	84.29	85.27	83.72	83.63	83.53	85.87	**85.95**	78.07	76.37	75.42	74.95	76.85	**79.63**
6	93.44	94.22	**94.67**	89.05	86.59	90.56	85.48	84.45	**94.63**	89.59	**85.82**	84.91	79.33	83.32	85.05
7	96.20	87.20	96.50	95.41	**100.00**	94.40	95.52	95.60	97.04	**98.65**	97.65	97.32	97.59	88.21	**97.98**
8	91.68	89.08	**93.76**	93.12	89.90	68.74	80.87	82.27	77.85	**82.89**	72.51	86.11	80.13	**86.13**	76.76
9	**95.52**	87.61	93.18	91.53	93.08	88.02	**88.10**	84.14	85.94	85.48	77.97	69.86	**83.02**	83.01	72.60
10	92.04	92.86	89.83	90.87	**93.12**	89.43	79.53	87.19	**90.40**	82.34	82.72	**85.06**	82.99	82.82	84.55
11	90.18	94.55	93.65	92.68	**98.90**	93.34	**96.21**	91.88	95.12	84.64	89.00	85.52	87.38	82.61	**90.04**
12	85.66	85.22	85.80	86.83	**93.48**	82.15	83.02	73.20	**88.70**	80.09	80.26	80.05	74.14	76.38	**83.47**
13	95.84	97.63	96.07	100.00	**100.00**	93.82	95.80	95.25	98.41	**100.00**	94.35	**97.12**	95.66	95.41	94.66
OA	91.49	91.55	91.50	**92.65**	91.85	87.84	87.78	86.93	**89.45**	87.81	84.87	84.33	83.99	**85.95**	84.91
(%)	(3.56)	(2.89)	(2.85)	**(2.18)**	(2.73)	(3.42)	(2.61)	(2.86)	**(3.68)**	(3.32)	(5.27)	(2.29)	(3.83)	**(4.37)**	(2.42)
AA	90.32	89.68	89.86	**90.77**	88.57	86.79	85.99	85.80	**87.94**	85.72	82.95	81.58	81.31	**83.02**	83.21
(%)	(2.38)	(3.26)	(2.34)	**(2.97)**	(3.45)	(3.65)	(2.55)	(2.67)	**(3.38)**	(3.57)	(4.68)	(2.41)	(3.78)	**(4.04)**	(2.00)
KAPPA	0.91	0.91	0.91	**0.92**	0.91	0.86	0.86	0.85	**0.88**	0.86	0.83	0.83	0.82	**0.84**	0.83
	(0.04)	(0.03)	(0.03)	**(0.02)**	(0.03)	(0.04)	(0.03)	(0.03)	**(0.04)**	(0.04)	(0.06)	(0.03)	(0.04)	**(0.04)**	(0.03)

I. Performance Evaluation

To demonstrate the effectiveness of the proposed AAN, we compared the AAN with multiple baselines on the KSC, UP, and SV datasets. Six methods are used as the compared baselines, including DPNLD [36], K-SDP [39], RLPA [37], 3D-CNN [35], SSRN [13], and DCRN [32]. Specifically, DPNLD, K-SDP, and RLPA are the data-cleaning methods,

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TABLE VIII

CLASSIFICATION ACCURACY FOR THE KSC DATASET. CLASSIFICATION ACCURACY OBTAINED BY THE AAN WITHOUT SHUFFLING, AND AAN, RESPECTIVELY, WITH EIGHT TRUE SAMPLES AND DIFFERENT NUMBER OF MISLABELED SAMPLES

class	The number of true and noisy training samples(KSC)					
	8(clean)+4(noisy)		8(clean)+8(noisy)		8(clean)+12(noisy)	
	w/o shuffle	shuffle	w/o shuffle	shuffle	w/o shuffle	shuffle
1	93.97	**97.36**	**95.02**	92.53	85.98	**97.64**
2	83.63	**89.15**	73.71	**92.29**	84.50	**86.09**
3	**97.24**	90.11	**94.33**	83.89	88.61	**91.33**
4	68.12	**78.69**	**69.29**	60.53	**57.48**	53.78
5	**89.15**	85.27	82.36	**85.95**	76.85	**78.99**
6	89.05	**89.60**	88.55	**94.63**	83.32	**85.69**
7	95.41	**98.60**	93.89	**97.04**	**97.01**	88.21
8	83.86	**93.12**	77.70	**77.85**	71.45	**86.13**
9	**96.71**	91.53	84.23	**85.94**	**83.60**	83.01
10	90.87	**93.94**	86.98	**90.40**	73.54	**82.82**
11	**97.05**	92.68	92.59	**95.12**	**88.67**	82.61
12	86.93	**87.61**	85.80	**88.70**	76.38	**79.46**
13	97.78	**100.00**	95.88	**98.41**	95.31	**95.41**
OA	91.95	**92.65**	88.10	**89.45**	83.82	**85.95**
(%)	(3.41)	**(2.18)**	(3.80)	**(3.68)**	(4.78)	**(4.37)**
AA	90.56	**90.77**	86.18	**87.94**	82.66	**83.02**
(%)	(3.46)	**(2.97)**	(4.09)	**(3.38)**	(4.55)	**(4.04)**
KAPPA	0.91	**0.92**	0.87	**0.88**	0.82	**0.84**
	(0.04)	**(0.02)**	(0.04)	**(0.04)**	(0.05)	**(0.04)**

and training data is cleaned by removing examples with labels that are likely to be tainted. Correspondingly, the 3D-CNN, SSRN, DCRN, and AAN are the deep-learning approaches by making a supervised learning architecture that is robust to noisy labels. All baselines are trained with identical data settings to ensure a fair comparison. For the KSC and SV, 24 clean samples per class are used for training, and the number of mislabeled samples per class are changed from 4 to 12. For the UP, 52 clean samples per class are used for training, and the number of mislabeled samples per class is changed from 8 to 24. The experimental results on KSC, SV, and UP are shown in Tables IX, X, and XI, respectively. It can be seen that deep-learning approaches outperform data-cleaning strategies. The reason is that data-cleaning methods may be impacted by the overcleaning issue, which excessively removes even clean samples. Consequently, the restricted training set is inadequate for learning the informative spatial and spectral representation.

In addition, the proposed AAN achieves the best performance among the deep-learning methods. However, the superiority of the AAN is not obvious when the number of noisy labels is small. One of the reasons is that simple challenges are insufficient for identifying robust models. The advantage of the proposed AAN will be more obvious with the increasing number of noisy labels. The phenomenon demonstrates the effectiveness of our method well. DCRN is a classical deep model for HSI classification with noisy labels. The unique characteristics of the AAN are as follows. One is that the ANRL function may achieve a finer tradeoff between NCE and RCE. Another is that the group-shuffle mechanism obtains a scattered spatial–spectral representation, which is a key factor in establishing a noise-robust deep network.

Fig. 3 visualizes the classification results produced by the compared methods that work on SV with 24 clean samples and the varied number of noisy labels (4, 8, and 12). Generally, the

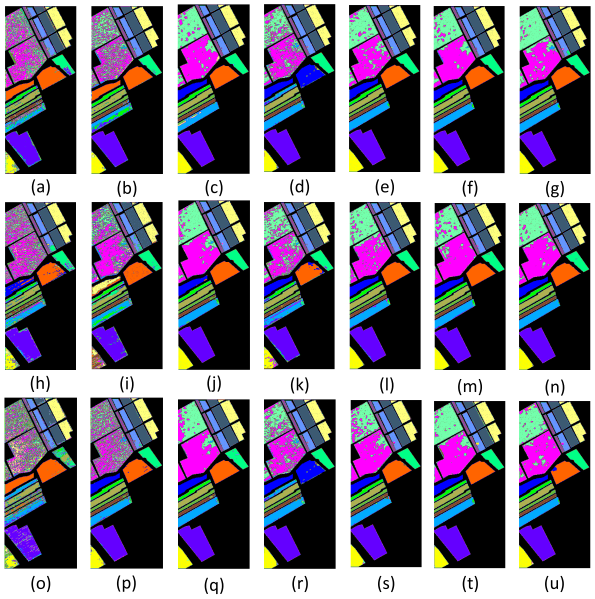

Fig. 3. Classification map for the Salinas Valley dataset. Classification map obtained by DPNLD (first column), K-SDP (second column), RLPA with extreme learning machine (ELM) classifier (third column), 3-D CNN (fourth column), SSRN (fifth column), DCRN (sixth column), and AAN (last column). (a)–(g) Four mislabeled samples for each class. (h)–(n) Eight mislabeled samples for each class. (o)–(u) 12 mislabeled samples for each class.

performance of models all declines as the proportion of noisy labels increases. The reason for this is that the noisy labels provide incorrect signals, resulting in inaccurate decisions and low generalization. The superiority of the proposed AAN can also be found in Fig. 3, which has clearer boundaries and neater regions than the other methods in all situations.

J. Performance Comparison With Limited Clean Samples

Experiments are also conducted to compare the performances of DPNLD, K-SDP, RLPA, 3-D CNN, SSRN, DCRN, and AAN with few clean samples in the training set. For the KSC and SV, eight clean samples per class are used for training and the number of mislabeled samples per class are 4, 8, and 12. For the UP, 20 clean samples per class are used for training and the number of mislabeled samples per class are 8, 16, and 24. The experimental results on the KSC, SV, and UP are shown in Fig. 4.

It is observable that all models perform worse since there are less clean examples available. For those data cleaning strategies (such as DPNLD, K-SDP, and RLPA), there is a greater degree of reduction in performance. The primary reason is that the number of samples utilized for training is significantly reduced during the noisy label removal step, which leaves the models vulnerable to underfitting. Additionally, neither the 3-D CNN nor the SSRN findings are ideal. The poor results are caused by the training set's complicated organizational structure, where the few clean examples make it harder for models with many parameters to understand the guiding signals. The AAN also achieves the best performance under the condition of limited training samples. AAN is a potent method for HSI classification with noisy labels as

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TABLE IX

CLASSIFICATION ACCURACY FOR THE KSC DATASET. CLASSIFICATION ACCURACY OBTAINED BY THE DPNLD, K-SDP, RLPA, 3-D CNN, SSRN, DCRN, AND AAN WITH 24 TRUE SAMPLES AND A DIFFERENT NUMBER OF MISLABELED SAMPLES

Metrics	The number of true and noisy training samples(KSC)																						
	24(clean)+4(noisy)							24(clean)+8(noisy)							24(clean)+12(noisy)								
	DPNLD	K-SDP	RLPA	CNN	SSRN	DCRN	**AAN**	DPNLD	K-SDP	RLPA	CNN	SSRN	DCRN	**AAN**	DPNLD	K-SDP	RLPA	CNN	SSRN	DCRN	**AAN**		
1	92.51	91.89	89.35	89.57	97.56	98.72	**99.48**	94.88	90.44	90.15	90.51	95.96	**99.23**	98.86	95.58	91.01	89.22	92.82	94.06	98.16	**99.20**		
2	80.10	81.37	85.66	83.52	95.20	97.69	**97.96**	77.01	78.25	78.66	71.49	96.55	97.69	**98.17**	68.39	85.83	70.73	77.44	96.04	94.22	**94.88**		
3	93.70	95.81	93.07	93.89	98.62	93.80	**99.35**	92.39	91.85	78.32	92.49	98.21	95.50	**97.82**	86.08	95.02	93.50	89.53	96.88	95.42	**97.96**		
4	57.14	48.99	58.05	46.91	82.80	**94.75**	94.30	47.27	53.65	59.39	51.47	83.47	81.97	**93.64**	52.92	48.81	50.66	53.70	82.87	79.85	**89.94**		
5	78.97	81.44	77.18	77.00	94.72	**94.24**	92.98	83.59	76.47	83.10	76.37	89.39	**97.28**	93.51	80.41	77.23	77.07	78.71	92.60	**95.40**	87.88		
6	83.11	85.28	82.58	74.86	98.77	98.56	**99.30**	76.67	87.47	77.32	81.30	91.79	97.10	**98.52**	81.92	84.79	85.02	67.14	89.29	94.10	**97.70**		
7	97.26	100.00	98.44	88.05	100.00	91.54	**98.84**	98.76	89.77	93.35	99.45	99.34	**100.00**	96.94	91.88	93.77	96.65	94.08	96.77	95.96	**97.20**		
8	65.71	93.47	73.63	81.75	94.73	**100.00**	99.68	78.65	76.65	72.94	82.34	97.91	98.91	**99.10**	79.79	84.27	71.71	86.34	93.82	**99.90**	99.67		
9	81.95	85.25	84.18	85.22	96.52	**99.94**	98.92	80.56	88.16	87.06	81.17	93.71	94.66	**98.87**	74.80	83.43	76.82	73.74	93.37	**99.51**	97.52		
10	78.59	89.16	71.83	79.82	98.69	98.49	**99.39**	76.00	87.77	85.32	73.49	91.20	**100.00**	98.28	80.10	85.71	84.56	81.64	95.99	99.17	98.66		
11	96.90	94.21	90.54	90.70	96.99	97.18	**99.82**	97.34	92.11	93.91	93.41	93.71	98.28	**99.07**	91.72	86.98	90.32	89.34	96.49	99.50	**99.92**		
12	76.06	81.91	79.94	79.06	94.64	97.15	**97.58**	82.99	79.14	82.29	81.66	91.68	**98.25**	97.06	75.41	75.54	79.93	73.90	92.47	95.95	**95.96**		
13	100.00	89.81	97.34	96.20	100.00	100.00	**100.00**	94.07	96.43	87.88	88.42	100.00	100.00	**100.00**	94.10	95.85	96.45	89.39	100.00	99.91	**100.00**		
OA	85.43	87.25	84.90	84.97	96.71	98.19	**98.86**	85.36	86.25	83.99	83.35	94.98	97.61	**98.38**	83.74	85.74	83.85	82.65	94.82	97.33	**97.86**		
(%)	(3.11)	(4.40)	(3.09)	(4.32)	(1.47)	(1.08)	**(0.67)**	(4.16)	(3.77)	(6.91)	(3.96)	(2.18)	(1.95)	**(0.55)**	(3.86)	(4.80)	(3.35)	(2.80)	(1.23)	(1.13)	**(0.60)**		
AA	83.23	86.04	83.21	82.04	96.10	97.08	**98.28**	83.09	83.70	82.28	81.81	94.07	96.84	**97.68**	81.01	83.71	81.74	80.60	93.90	95.93	**96.65**		
(%)	(3.28)	(2.74)	(2.30)	(3.56)	(1.86)	(2.38)	**(0.73)**	(3.89)	(3.32)	(5.14)	(3.95)	(1.98)	(2.73)	**(0.91)**	(3.73)	(4.46)	(3.27)	(3.24)	(1.82)	(2.37)	**(0.89)**		
KAPPA	0.84	0.86	0.83	0.83	0.96	0.98	**0.99**	0.84	0.85	0.82	0.81	0.94	0.97	**0.98**	0.82	0.84	0.82	0.81	0.94	0.97	**0.98**		
	(0.03)	(0.05)	(0.03)	(0.05)	(0.02)	(0.01)	**(0.01)**	(0.05)	(0.04)	(0.08)	(0.04)	(0.02)	(0.02)	**(0.01)**	(0.04)	(0.05)	(0.04)	(0.03)	(0.01)	(0.01)	**(0.01)**		

TABLE X

CLASSIFICATION ACCURACY FOR THE SALINAS DATASET. CLASSIFICATION ACCURACY OBTAINED BY THE DPNLD, K-SDP, RLPA, 3-D CNN, SSRN, DCRN, AND AAN WITH EIGHT TRUE SAMPLES AND A DIFFERENT NUMBER OF MISLABELED SAMPLES

Metrics	The number of true and noisy training samples(salinas)																						
	24(clean)+4(noisy)							24(clean)+8(noisy)							24(clean)+12(noisy)								
	DPNLD	K-SDP	RLPA	CNN	SSRN	DCRN	**AAN**	DPNLD	K-SDP	RLPA	CNN	SSRN	DCRN	**AAN**	DPNLD	K-SDP	RLPA	CNN	SSRN	DCRN	**AAN**		
1	93.87	98.81	98.53	81.92	99.64	**99.40**	96.22	96.48	91.76	97.37	77.96	99.74	99.69	**99.98**	97.70	87.82	99.42	85.32	99.74	97.82	**98.48**		
2	98.56	93.13	99.15	99.41	99.92	100.00	**100.00**	99.78	99.96	99.77	86.74	99.91	98.98	**99.99**	97.59	99.87	98.06	95.51	99.84	90.17	**98.63**		
3	88.18	90.64	95.43	88.30	100.00	99.48	**99.58**	79.74	77.02	92.66	76.62	99.44	**99.91**	97.50	80.09	86.90	96.51	71.12	97.17	**99.96**	98.10		
4	98.30	97.76	99.67	91.64	99.07	99.58	**99.65**	92.95	97.18	97.68	85.15	99.40	99.82	**99.10**	86.93	89.78	98.59	96.49	99.36	89.17	**98.64**		
5	91.73	90.90	97.13	73.54	97.20	96.83	**98.62**	90.19	87.13	89.79	90.32	96.75	97.62	**97.84**	83.54	93.42	95.21	74.48	93.65	**97.80**	94.02		
6	96.01	98.42	96.93	94.13	100.00	**99.84**	99.59	93.42	97.76	98.74	96.88	99.71	99.68	**100.00**	96.55	98.83	97.58	93.83	98.95	99.32	**99.49**		
7	94.66	99.20	99.17	95.71	99.69	**99.91**	99.88	91.42	97.52	98.24	85.07	99.75	99.44	**99.95**	95.49	96.81	99.77	93.89	99.55	**99.66**	99.23		
8	59.38	65.74	76.19	43.17	78.56	78.86	**82.57**	66.62	57.51	75.12	47.27	73.77	81.92	**81.90**	56.81	55.39	60.57	54.52	71.52	**83.38**	78.77		
9	96.65	97.61	99.40	90.89	99.86	99.48	**99.81**	98.49	98.28	99.31	89.30	99.71	**100.00**	99.99	98.39	97.95	99.00	79.48	99.10	99.42	**99.96**		
10	76.15	77.13	89.23	49.92	95.28	**98.03**	96.73	71.81	67.73	82.65	60.54	96.03	**94.24**	93.16	67.49	76.54	89.98	62.46	83.91	87.52	**95.79**		
11	97.27	96.24	93.29	83.66	99.64	99.34	**99.88**	89.83	93.92	93.09	88.70	97.97	99.57	**99.68**	91.66	89.34	93.14	93.16	97.67	**99.46**	99.33		
12	95.43	95.08	98.49	91.36	99.71	99.89	**99.98**	93.55	99.04	99.79	80.30	99.91	**99.98**	99.22	87.52	88.32	98.61	86.48	98.92	99.36	**99.89**		
13	95.48	97.73	98.16	84.11	99.52	99.43	**99.83**	98.65	92.42	98.16	84.53	99.85	**99.50**	98.47	93.29	95.22	98.78	94.03	97.84	98.94	**99.59**		
14	93.64	92.86	96.96	84.89	97.86	98.80	**99.06**	84.37	92.44	91.45	94.18	97.40	98.43	**99.15**	89.34	85.43	96.42	85.49	95.57	**98.57**	96.83		
15	71.96	70.55	72.57	76.16	82.30	88.32	89.57	69.74	79.88	76.30	64.91	89.12	79.45	**86.45**	78.82	80.04	87.31	53.24	73.87	79.95	**88.45**		
16	89.78	84.91	94.04	80.38	97.78	98.25	**99.25**	81.50	85.69	86.77	82.64	97.36	**99.34**	97.60	82.97	83.45	89.02	77.46	89.61	95.06	**98.70**		
OA	83.30	84.76	89.44	75.60	92.47	93.75	**94.14**	83.10	83.17	88.55	74.09	92.32	92.75	**93.58**	81.80	82.97	87.95	73.68	88.35	91.53	**92.96**		
(%)	(4.00)	(4.70)	(2.37)	(4.99)	(2.48)	(2.25)	**(0.66)**	(5.01)	(4.69)	(2.57)	(3.53)	(3.74)	(1.54)	**(0.88)**	(5.07)	(4.54)	(3.70)	(4.73)	(2.60)	(3.68)	**(1.32)**		
AA	89.82	90.42	94.02	81.82	96.63	97.43	**97.50**	87.41	88.45	92.31	80.69	96.61	96.72	**96.87**	86.51	87.82	93.62	81.06	93.52	94.72	**96.50**		
(%)	(1.60)	(2.76)	(1.53)	(3.70)	(0.76)	(0.75)	**(0.69)**	(3.97)	(3.10)	(2.00)	(3.26)	(0.99)	(0.78)	**(0.71)**	(4.57)	(3.05)	(1.89)	(4.50)	(1.73)	(3.11)	**(0.86)**		
KAPPA	0.82	0.83	0.88	0.73	0.92	0.93	**0.94**	0.81	0.81	0.87	0.71	0.91	0.92	**0.93**	0.80	0.81	0.87	0.71	0.87	0.91	**0.92**		
	(0.04)	(0.05)	(0.03)	(0.05)	(0.03)	(0.02)	**(0.01)**	(0.05)	(0.05)	(0.03)	(0.04)	(0.04)	(0.02)	**(0.01)**	(0.06)	(0.05)	(0.04)	(0.05)	(0.03)	(0.04)	**(0.01)**		

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TABLE XI

CLASSIFICATION ACCURACY FOR THE UP DATASET. CLASSIFICATION ACCURACY OBTAINED BY THE DPNLD, K-SDP, RLPA, 3-D CNN, SSRN, DCRN, AND AAN WITH 52 TRUE SAMPLES AND A DIFFERENT NUMBER OF MISLABELED SAMPLES

Metrics	The number of true and noisy training samples(UP)																						
	52(clean)+8(noisy)							52(clean)+16(noisy)							52(clean)+24(noisy)								
	DPNLD	K-SDP	RLPA	CNN	SSRN	DCRN	**AAN**	DPNLD	K-SDP	RLPA	CNN	SSRN	DCRN	**AAN**	DPNLD	K-SDP	RLPA	CNN	SSRN	DCRN	**AAN**		
1	67.86	76.82	88.57	83.55	91.21	**97.69**	96.88	75.59	78.62	90.57	62.58	87.73	**95.98**	94.76	70.41	78.57	82.71	71.07	84.80	93.86	**94.05**		
2	71.54	77.05	85.49	65.57	93.97	96.53	**98.79**	67.91	73.80	81.34	73.93	89.11	95.85	**97.04**	64.13	70.39	82.52	70.03	87.07	92.82	**96.06**		
3	79.53	83.01	80.61	80.52	89.86	91.18	**95.80**	76.52	73.19	87.15	75.93	84.66	90.19	**94.71**	72.2	76.59	84.07	73.87	88.11	**88.63**	83.35		
4	82.07	81.93	91.65	82.59	96.14	97.13	**97.78**	80.11	81.54	91.08	83.61	92.78	96.91	**96.93**	75.01	86.00	92.10	79.38	92.00	95.03	**95.86**		
5	99.72	96.99	99.41	98.68	99.87	99.35	**99.98**	99.03	90.04	99.89	96.93	99.90	99.88	**99.93**	89.51	93.09	99.54	95.00	99.58	99.36	**99.77**		
6	87.84	87.16	98.47	84.52	98.20	**99.60**	98.13	80.12	83.50	94.41	79.79	94.98	**98.48**	97.84	76.78	80.34	86.82	83.46	98.35	96.66	**97.49**		
7	90.92	94.13	99.03	87.99	98.88	99.73	**99.75**	88.85	88.91	97.46	89.23	97.20	97.69	**99.50**	84.94	96.00	91.41	93.91	97.93	**99.45**	98.74		
8	72.84	63.34	86.63	72.30	92.42	**95.57**	93.59	80.43	78.77	84.26	75.53	89.84	**92.81**	92.03	70.81	76.82	82.53	70.82	85.91	87.37	**94.21**		
9	95.04	95.61	98.91	98.11	99.69	99.32	**99.65**	93.87	94.68	99.6	97.1	98.49	99.25	**99.59**	96.34	94.68	97.41	93.83	98.26	99.59	**99.73**		
OA	76.09	79.19	88.93	75.50	94.30	97.08	**97.76**	75.05	78.04	87.00	75.44	90.48	95.97	**96.44**	70.43	76.79	84.93	74.68	89.31	93.62	**95.38**		
(%)	(4.75)	(4.10)	(2.53)	(4.98)	(2.18)	(0.76)	**(0.75)**	(6.19)	(5.94)	(3.00)	(5.93)	(2.58)	(1.06)	**(1.03)**	(8.35)	(5.20)	(3.35)	(6.78)	(3.24)	(3.08)	**(1.44)**		
AA	83.04	84.00	92.09	83.76	95.58	97.49	**97.65**	82.49	82.56	91.75	81.63	92.74	96.34	**96.91**	77.79	83.61	88.79	81.26	92.45	95.03	**95.41**		
(%)	(2.23)	(2.87)	(1.47)	(3.77)	(1.41)	(0.50)	**(0.41)**	(5.11)	(4.08)	(1.46)	(4.36)	(1.92)	(0.89)	**(0.80)**	(6.23)	(3.40)	(1.55)	(4.03)	(2.13)	(1.30)	**(1.09)**		
KAPPA	0.70	0.74	0.86	0.70	0.93	0.96	**0.97**	0.69	0.72	0.83	0.69	0.88	0.95	**0.95**	0.64	0.71	0.81	0.69	0.86	0.92	**0.94**		
	(0.05)	(0.05)	(0.03)	(0.06)	(0.03)	(0.01)	**(0.01)**	(0.07)	(0.07)	(0.04)	(0.07)	(0.03)	(0.01)	**(0.01)**	(0.09)	(0.06)	(0.04)	(0.07)	(0.04)	(0.04)	**(0.02)**		

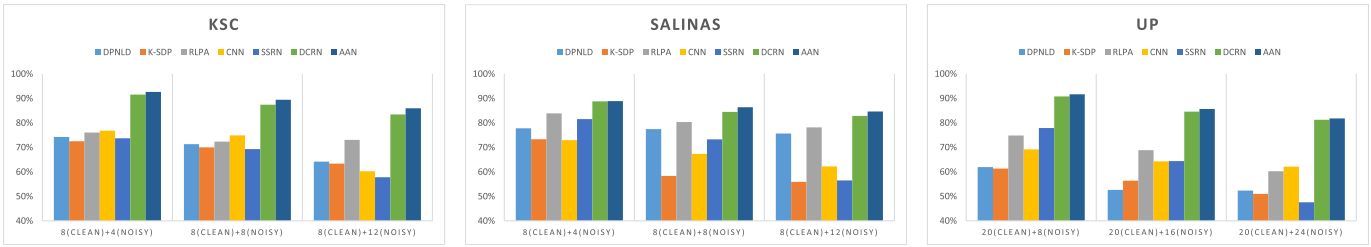

Fig. 4. OA comparisons with few clean samples.

compared to DCRN due to its scatter spatial–spectral feature representation and superior robust loss function.

V. DISCUSSION

A. "Spectral and Spatial" or "Spectral–Spatial"?

We propose a spatial–spectral network to address the problem of HSI classification in the presence of noisy labels in the training set. The proposed method is developed from DCRN [32], which is the first attempt to introduce an end-to-end deep neural network to deal with noisy labels for HSI classification. DCRN avoids noise-cleansing errors and outperforms traditional data-cleaning methods. However, DCRN combines the spatial and spectral branches separately, where the spatial and spectral features are not exploited adequately. Compared to DCRN, we propose a spatial–spectral method for robust HSI classification. The experiment demonstrates that the AAN can achieve superior results than DCRN.

B. "Attention" or "not Attention"?

We introduce an attention block to a noise-robust deep architecture to give attention to the clean or noisy training set at the same time. The attention mechanism has been demonstrated as a powerful tool for HSIc when the training set is clean. However, as illustrated in Section IV-F, the regular attention block shows helplessness against noisy labels in the training set. To facilitate practical HSI applications, we accommodate the attention mechanism to a noise-robust deep network with the help of group-shuffle operation. In this manner, the scattered spatial–spectral features of interest are obtained to alleviate the interlocking of deep models. The experiment illustrates that group-shuffle attention is suited for practical HSI applications, regardless of whether noisy labels are present in the training set or not.

C. "RCE" or "NCE"?

We develop an ANRL function to prevent overfitting for the classifier learned with noisy labels. RCE and NCE are two mutually boosted robust loss functions that are often tied together for robust training. However, it is difficult to determine which one is more important when the noise ratios are unpredictable. In this article, we explore an adaptive way by learning a balance parameter to decide which one does more. Experimental results show that the importance of RCE increases as the number of noisy labels grows. Experiments also demonstrate that the adaptive manner to balance NCE and RCE is better than the averaged way for noise-robust models.

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VI. Conclusion

This article proposes AAN, a novel robust network to deal with noisy labels in the training data for HSI classification. This article accommodates the attention mechanism to a noise-robust deep architecture for the first time by utilizing the group-shuffle operations to obtain robust spatial–spectral representation in the presence of noisy labels. A novel ANRL function is also designed to adapt to different proportions of noisy labels. We demonstrate the state-of-the-art performance of our method with extensive experiments on three HSI datasets with noisy labels. The future direction of our work is to exploit a self-supervised method to deal with noisy labels.

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